

14 MASTER STUDY ORDER: what to do, in what order

Built 16 Sep 2026. This is THE order document. It ties together everything we hold: the notes, the slide decks, the reference cards, the patterns, the drill, the papers. Follow it top to bottom.

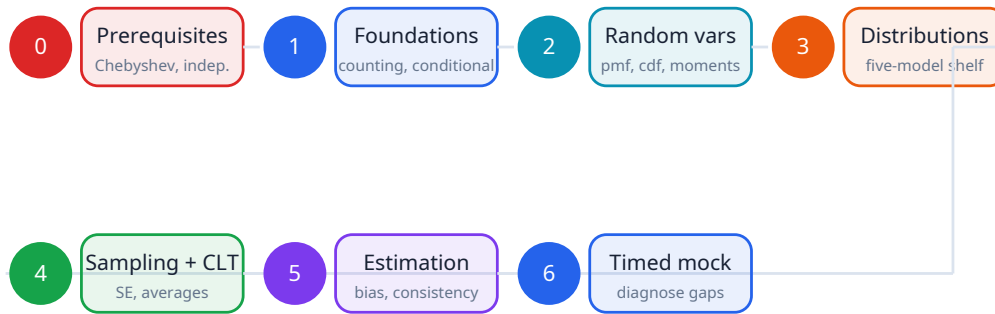
Supersedes [reports/11-QUESTION-ATLAS/00-LEARN-ORDER.md](#) (that one was written before the notes folder existed; this one covers every artifact we now have).

0. THE ARTIFACT INVENTORY (what exists, where)

A. THE NOTES	deck/notes/	(11 files)
the slides rewritten as teachable notes. READ THESE FIRST.		
you do NOT need to open a PDF to learn the theory.		
B. THE DECKS	the S&P batch-2 decks (text copies)	
L1-7, L8-9, L10-11 Cheb, L12-13, L14-15		
these are PASS 2 / the visual check, NOT new teaching		
(except L10-11 Chebyshev, which IS new)		
C. THE REFERENCE	deck/*.md	(15 files)
cheatsheet, terms, linguistics, methods, distributions, numbers,		
patterns, traps, questions, slide map, hidden layer, drill,		
mechanics. USE THESE WHILE SOLVING, not as reading material.		
D. THE PATTERNS	deck/07-PATTERNS.md	
the mutation system. read AFTER two papers, then it pays off.		
E. THE PAPERS	md/paper-mte-*, paper-ete-*, resess-*, asgn-*	
the drill material. papers FIRST, then assignments.		
F. THE ORIGINAL SLIDES the PDFs + md/ conversions		
ONLY open these for (a) Chebyshev deck, (b) L1-7/L8-9/L12-13/		
L14-15 question blocks. Everything else is in the notes already.		

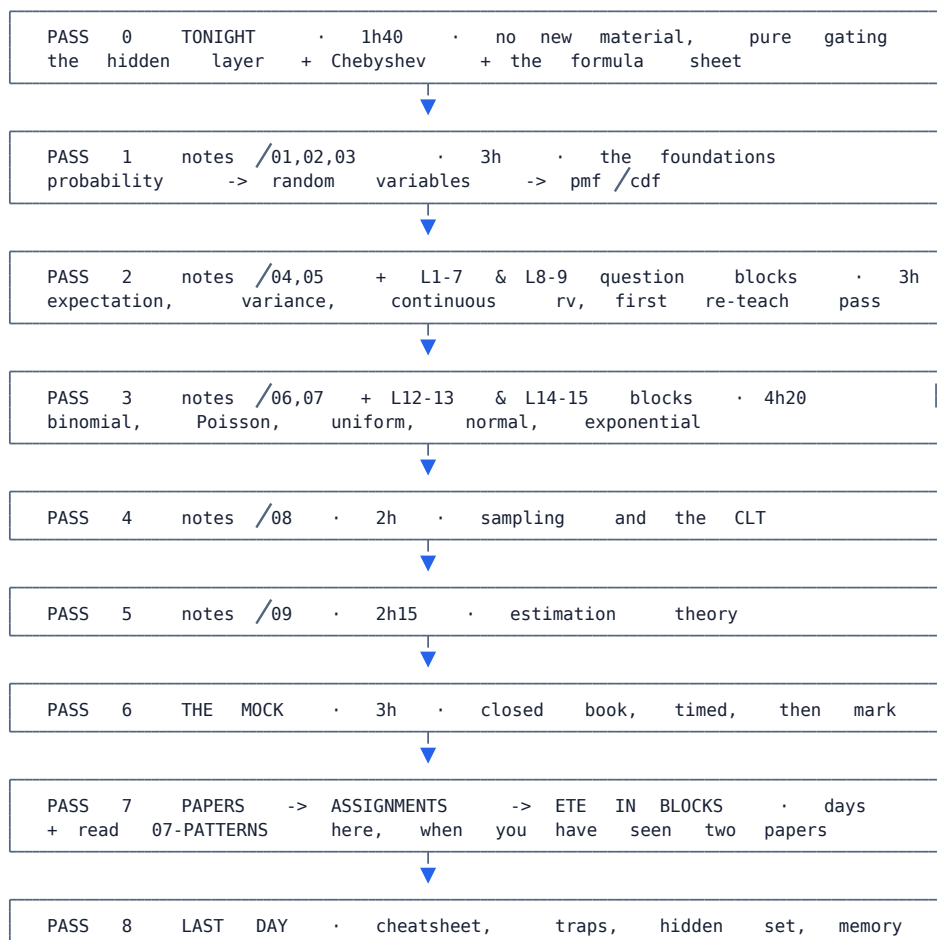
1. THE MASTER ORDER, ONE PICTURE

Study order with dependency gates



advance only after the gate for the current pass is met

Study order and dependency gates



2. PASS 0, TONIGHT, 1h40, THE GATING BLOCK

Do this FIRST. It gates four downstream topics and carries marks in every paper.

#	WHAT	TIME
0.1 notes/10 PART A: Chebyshev, all of it	30m	
	(skip Q3, errata 15; do Q1 and Q2)	
0.2 notes/10 PART B: H1 independence rules	25m	
	$E(XY)$, $\text{Var}(X+Y)$, $\text{Var}(X-Y)$, then the drill:	
	35/12 -> 35/6 -> 35/54	
0.3 deck/11-BEYOND-SLIDES.md, H3 to H7	30m	
	memoryless, landmarks, $N \times P$, $1/p$, nK/N	
0.4 deck/01-CHEATSHEET.md, read aloud, twice	15m	
GATE: write all of H1-H7 on blank paper, no notes.		
ALSO: the Chebyshev deck's own pages, 9 pages, text at		
~/mas2001-devore/corpus-text/L10-11-Cheb.txt		

Why first: [07 - PATTERNS.md](#) shows Chebyshev is graded 9+ times; H1 gates the dice variance, $\text{Var}(\bar{X})$, every estimator variance and Poisson additivity. Doing this cold costs 1h40 now and saves rework in PASS 3, 4 and 5.

3. PASS 1, THE FOUNDATIONS, 3h

1.1	notes/01-probability-foundations.md	55m
	read fully, then DO the 3 worked examples	
	(tournament, laptops, basketball) on paper	
1.2	notes/02-random-variables.md	45m
	the 36-cell dice grid, draw it yourself	
	+ deck/05-DISTRIBUTIONS.md first table	
1.3	notes/03-pmf-and-cdf.md	60m
	DO: gas pumps, find-k (the $10k^2+9k=1$ one),	
	boards inspection, dice maximum M, the proof	
1.4	deck/04-METHODS.md sections 1-5	20m
	the solving paths for what you just read	

4. PASS 2, EXPECTATION + VARIANCE + FIRST RE-TEACH, 3h15

2.1 notes/04-expectation-and-variance.md	55m	
	DO: the die (3.5), the p094 triple (209),	
	the freezer 6-part chain, the magazine 3-vs-4	
2.2 notes/05-continuous-rv.md	65m	
	DO: $f=2x$, the conditional 5/12, the bus	
	triangle 6 parts, cdf->pdf, the hospital Y	
2.3 S&P L1-7 QUESTIONS ONLY (13 blocks, pass 2)	45m	
	text: ~/mas2001-devore/corpus-text/L1-7.txt	
	these are the SAME skeletons as notes/01-03	
2.4 S&P L8-9 QUESTIONS ONLY (4 blocks)	30m	
	bus pdf, CDF-to-density, kidney $E(Y)$, Pareto	
GATE: the freezer chain end to end. The L1-7 questions solved		
WITHOUT looking at the notes/01-03 answers.		
NOTE: do NOT read L1-7/L8-9 as new teaching. 54-55% of their		
5-grams exist in the notes deck already. They are pass 2.		

5. PASS 3, THE DISTRIBUTION SHELF, 4h20

One template per distribution: setup, pmf/pdf, moments, the corpus question, the trap.

3.1	notes/06-binomial-poisson.md PART A binomial	55m
	DO: pens 3 parts, irregular die $p=5/8$,	
	5-coin derivation	
	then L12-13 binomial block (pass 2)	
3.2	notes/06 PART B Poisson	50m
	DO: ratio $\lambda=10$, THE NESTING 0.00145,	
	insurance 0.1755, the tail 0.8753	
3.3	notes/07 PART A uniform	25m
	DO: $U(2,6)$, the length ratio 0.50	
3.4	notes/07 PART B normal	70m
	DO: the 8.6 chain, the 45-62 interval, the	
	20% inverse (3.792), the two-unknown (errata	
	10: μ 37.2, σ 28.2, cutoff 30.4)	
	then L14-15 normals (pass 2)	
3.5	notes/07 PART C exponential	40m
	DO: 15/hr under 3 min (0.5276), the	
	conditional 0.6225, memoryless, unit conv	
	then L14-15 exponential (pass 2)	
3.6	deck/05-DISTRIBUTIONS.md, the 9-slot grid	20m
	mark which slots are ASKED vs a SIBLING	

6. PASS 4, SAMPLING AND CLT, 1h40

4.1	notes/08-sampling-and-clt.md, read fully	60m
	DO: lightbulbs (SE 20/10), ATM (0.0668),	
	impurity ($Z_2 = -0.94$, 0.1645 table), LED	
	machines $n=9$ (0.6898)	
4.2	deck/04-METHODS.md sections 14-15	15m
	the SE and CLT solving paths	
4.3	redo the machines $n=9$ case BLIND, then check	25m
	(it is the template: normal pop, small n)	

7. PASS 5, ESTIMATION, 2h15

5.1	notes/09-estimation.md, read fully	70m
	DO BOTH comparison sets on paper, all steps:	
	set 1 -> $\lambda=0$, t_1/t_3 unbiased, t_1 wins	
	set 2 -> $\lambda=1$, all unbiased, T_3 wins	
	+ the consistency trap (fixed n is NOT it)	
5.2	the 3 real-life numerics	25m
	205 ms (state estimator AND estimate),	
	0.93, the battery CI (7.81, 8.99) [boundary]	
5.3	lms-theory p024-p025 ONLY (sufficiency)	25m
	SKIP p031-p040 entirely, out of MTE	
5.4	deck/04-METHODS.md section 16	15m
	the estimator-comparison protocol	

8. PASS 6, THE MOCK, 3h

6.1 reports/07-MOCK-PAPER.md, CLOSED BOOK, TIMED	90m	
6.2 mark it against reports/08-MOCK-SOLUTIONS.md	30m	
	(errata 18: the B4 note is wrong, T4 is	
	THIRD of four, not smallest)	
6.3 repair ONLY the misses, then redo those parts	40m	
6.4 deck/08-TRAPS.md, the 10-item list, aloud	20m	
GATE:		
24+/30	-> proceed to PASS 7	
18-23	-> the next morning goes to the WEAKEST pass only	
<18	-> the next morning re-reads notes/01-05 (the base)	

9. PASS 7, PAPERS AND ASSIGNMENTS, several days

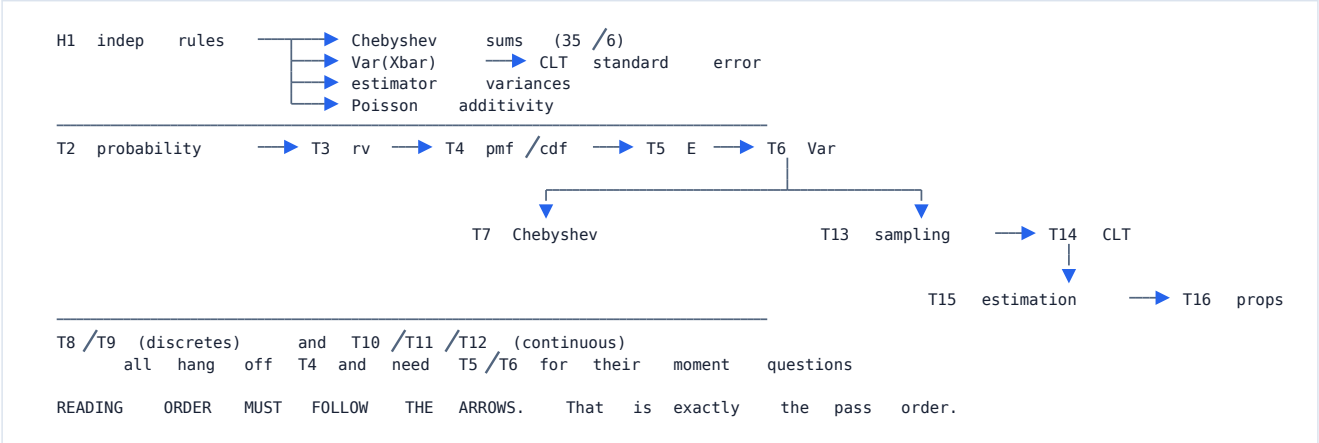
Order matters: papers reveal the SHAPE, assignments give the VOLUME.

7.1	MTE 2025-26 paper, closed book, then mark	60m
	md/paper-mte-2025-26.md + the scheme	
7.2	MTE 2024-25 paper, same	60m
7.3	READ deck/07-PATTERNS.md NOW	30m
	you have seen two papers, so the mutation	
	system will click. Then re-scan both papers	
	and label every question with its RADICAL.	
7.4	deck/09-QUESTIONS.md, the tagged inventory	20m
	confirm your labels against it	
7.5	2025-26 assignment bundle, assignments 1 and 2	2h
	(the in-scope part). The full bundle has 119	
	rows; assignments 3-5 get a separate light pass.	
7.6	2024-25 assignments 1 and 2 (in MTE scope)	1h30
7.7	2024-25 assignments 3, 3-ep2, 4, 5 (LIGHT)	45m
7.8	ETE S3 and S4, IN blocks only (56 of 97)	2h
	where the same skeletons reappear under	
	exam wording	
7.9	re-sess papers, IN blocks only	40m
7.10	asgn-faculty-variant: DEDUP ONLY, do not drill	10m
	its key ($k=1/10$) contradicts its row (0.0782)	

10. PASS 8, THE LAST DAY

8.1	deck/01-CHEATSHEET.md, twice, aloud	40m
8.2	every mean/variance pair, from memory	15m
	+ the three Chebyshev values (3/4, 8/9, 15/16)	
8.3	the hidden set H1-H7, from memory	15m
8.4	deck/08-TRAPS.md, the 10-item list	10m
8.5	deck/13-EXAM-MECHANICS.md, the time budget	10m

11. THE DEPENDENCY GRAPH (why this order)



12. THE ARTIFACT-TO-PASS TABLE (one lookup)

ARTIFACT	PASS	ROLE
notes/10 (Chebyshev + hidden)	0	GATING, read first
deck/11-BEYOND-SLIDES.md	0	GATING
deck/01-CHEATSHEET.md	0,8	read aloud
notes/01 probability	1	theory
notes/02 random variables	1	theory
notes/03 pmf/cdf	1	theory
notes/04 expectation/variance	2	theory
notes/05 continuous rv	2	theory
L1-7 questions	2	pass 2
L8-9 questions	2	pass 2
notes/06 binomial/Poisson	3	theory
notes/07 uniform/normal/exp	3	theory
L12-13, L14-15 questions	3	pass 2
notes/08 sampling/CLT	4	theory
notes/09 estimation	5	theory
lms-theory p024-025	5	2 unique items only
mock + solutions	6	integration gate
MTE papers + schemes	7	drill, shape first
deck/07-PATTERNS.md	7	after 2 papers
deck/09-QUESTIONS.md	7	label your radicals
assignments (bundle, 2024-25)	7	drill, volume
ETE + re-sess IN blocks	7	drill under wording
deck/12-DRILL.md	any	as a ladder, 7 rungs
deck/04-METHODS.md	while	lookup while solving
deck/08-TRAPS.md	while	lookup
deck/02-TERMS.md	while	lookup
deck/03-LINGUISTICS.md	while	lookup, before drills
deck/06-NUMBERS.md	while	lookup
deck/05-DISTRIBUTIONS.md	while	lookup
deck/10-SLIDE-MAP.md	while	only if you open PDFs
deck/13-EXAM-MECHANICS.md	8	night before
lms-MLE, lms-MoM, handout	NEVER	out of scope
lms-theory p031-p040	NEVER	out of scope

13. IF TIME IS SHORT (fallbacks, in priority order)

ONLY 1 DAY (7h20 core, plus 40m review reserve):
 PASS 0 (1h40) + notes /01-03 skim (1h) + notes /06,07 skim (2h)
 + mock (1h30) + cheatsheet (40m) + traps (30m)
 -> the hidden layer + the shelf + the mock. Highest marks /hour.

ONLY 1 EVENING (3h40):
 PASS 0 (1h40) + the two MTE papers, closed book, then mark (2h)
 -> you will at least recognise every question shape.

ONLY 1 HOUR:
 deck /01-CHEATSHEET.md read aloud, twice.
 + the five distributions' mean /variance from memory.
 -> the single highest-value hour available.

14. RULES THAT MAKE THIS ORDER WORK

R1	hidden prerequisites FIRST. they gate 4 downstream topics. 1h40.
R2	read a topic's NOTES before its S&P re-teach deck.
R3	never read both of a pair cold:
(notes deck, L1-7)	(notes deck, L8-9)
(ppt3, L12-13)	(ppt4, L14-15)
R4	only L10-11 goes first, nothing else teaches Chebyshev.
R5	every question is done WITH its deck or paper, never read in a batch.
R6	papers BEFORE assignments (shape first, volume second).
R7	out-of-scope gets ONE light pass, never more.
R8	read deck/08-TRAPS.md before opening raw source PDFs. The guided question
blocks in PASS 2 and PASS 3 may come earlier.	
R9	the mock is the only timed closed-book artifact; do not re-run it
for comfort, only the failed parts.	
R10 the notes replace the PDFs for learning. Open the PDFs only for the	
Chebyshev deck and the four S&P question blocks.	

15. THE DAILY SHAPE (once you are in PASS 1+)

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09:00 read one notes / chapter end to end, no solving
10:00 solve every worked example in it on PAPER
11:00 the pass-2 deck block (if the pass has one)
12:00 break
13:00 deck /12-DRILL.md, one rung
14:00 mistakes: go to deck /04-METHODS for the slot
15:00 break
16:00 the pass gate (blank paper, no notes)
17:00 done. No new material after this.
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do NOT read two chapters without solving between them.
 the solve step is where the marks come from.

Master study order v1, 16 Sep 2026. Time estimates computed from the page counts and the worked-example density in each notes file. The errors/lies fix and the deep audit still await the user's

go.